Explainable AI Multi-Modal Decision Support System for Sugarcane Disease

Project Defense & Executive Presentation Deck

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Slide 2: The Agricultural Pathology Crisis & AI Solution

The Challenge: Sugarcane sustains 50M+ growers in India, but diseases like Red Rot, Brown Rust, Mosaic, and Yellow Leaf cause 15% to 100% crop loss. Standard AI models act as black boxes, provide no early warnings before foliar damage occurs, and frequently fabricate heuristic risk scores.

The Multi-Modal Solution: A unified dual-track architecture combining deep vision (EfficientNetV2-S + Grad-CAM++) for transparent leaf diagnosis and biometeorology (XGBoost + TreeSHAP + ALE) for 7-day advance early warning.

Slide 3: Vision Benchmark & Decision Gate Verdict

Model Benchmark: Whole-Image EfficientNetV2-S achieved **81.47% Macro-F1** on held-out test data and **74.56% Macro-F1** on 1,488 independent real-world field images.

Decision Gate: Evaluating a YOLO11 leaf-detection and cropping pipeline revealed that bounding box cropping sliced elongated red rot midrib lesions and fragmented dense field foliage, causing field accuracy to drop by **7.34%** while increasing CPU latency by 30%. YOLO11 was formally rejected; whole-image classification with Grad-CAM++ was retained.

Slide 4: Agrometeorological Risk Model & Ethical Transparency

Ethical Stance: Grounded on real RPCAU Pusa empirical field records (R^2 = 0.8186), the multi-station machine learning model is strictly designated as an **Experimental Agrometeorological Prototype** predicting microclimatic conduciveness (ICAR-SBI parameters), never as an uncalibrated biological disease occurrence ledger.

Performance: Tuned XGBoost achieved **98.72% Accuracy**, **0.9995 AUROC**, and **0.0093 Brier Score** (-77% calibration error vs. linear baseline), maintaining 95.86% accuracy when transferred out-of-region to Coimbatore, Tamil Nadu.

Slide 5: Rural Resiliency, Offline ONNX & Bilingual Impact

Rural Accessibility: Integrated bilingual interface (English and Tamil) with browser-native Web Speech TTS audio synthesis.

Offline Readiness: Exported ONNX model executes on CPU in **18.88 ms** with 5.36×10^{-7} max discrepancy compared to PyTorch.

Pre-Inference Quality Gate: OpenCV Laplacian blur detection (<60 variance) and Shannon entropy (>1.80 bits) uncertainty gates prevent overconfident misdiagnoses on poor-quality leaf photography.